2D Heat Convection, Steady State (Finite Differences)

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \varnothing \rightarrow \nabla^2 T = \varnothing \quad (1) \rightsquigarrow \text{Laplacian of Temperature}$$

$$\frac{1}{dx^2} T(i+1,j) - \frac{2}{dx^2} T(i,j) + \frac{1}{dx^2} T(i-1,j) +$$

$$\frac{1}{dy^2} T(i,j+1) - \frac{2}{dx^2} T(i,j) + \frac{1}{dy^2} T(i,j-1) = \varnothing \Rightarrow$$

$$\frac{1}{dx^2} T(i+1,j) + \left(-\frac{2}{dx^2} - \frac{2}{dy^2}\right) T(i,j) + \frac{1}{dx^2} T(i-1,j) +$$

$$\frac{1}{dy^2} T(i,j+1) + \frac{1}{dy^2} T(i,j-1) = \varnothing \Rightarrow$$

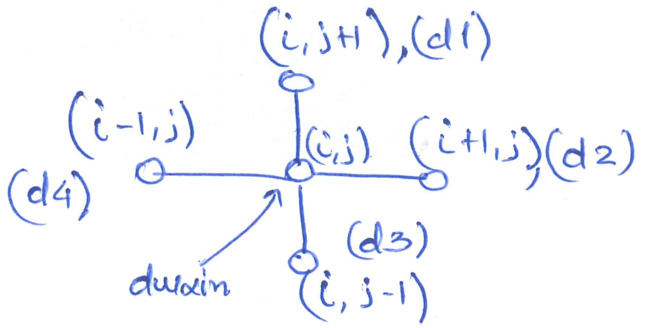
$$\frac{1}{dx^2} T(i+1,j) - \left(\frac{2}{dx^2} + \frac{2}{dy^2}\right) T(i,j) + \frac{1}{dx^2} T(i-1,j) +$$

$$\frac{1}{dy^2} T(i,j+1) + \frac{1}{dy^2} T(i,j-1) = \varnothing \Rightarrow$$

$$-\frac{1}{dx^2} T(i+1,j) + \left(\frac{2}{dx^2} + \frac{2}{dy^2}\right) T(i,j) - \frac{1}{dx^2} T(i-1,j) -$$

$$\frac{1}{dy^2} T(i,j+1) - \frac{1}{dy^2} T(i,j-1) = \varnothing \quad (2) \rightsquigarrow \underset{\sim}{A}\vec{x} = \vec{b}$$

Node Stencil



$$d_{main} = \frac{2}{dx^2} + \frac{2}{dy^2}$$

$$d1 = -\frac{1}{dy^2}, \quad d2 = -\frac{1}{dx^2}$$

$$d3 = -\frac{1}{dy^2}, \quad d4 = -\frac{1}{dx^2}$$

$$d2\, T(i+1,j) + d_{main}\, T(i,j) + d4\, T(i-1,j) +$$

$$d1\, T(i,j+1) + d3\, T(i,j-1) = \varnothing \quad (3)$$

$$d_{main}\, T(i,j) = -d1\, T(i,j+1) - d2\, T(i+1,j) - d3\, T(i,j-1) - d4\, T(i-1,j) \Rightarrow$$

$$T(i,j) = \frac{1}{d_{main}} \left[-d1\, T(i,j+1) - d2\, T(i+1,j) - d3\, T(i,j-1) - d4\, T(i-1,j)\right] \quad (4)$$

$$d_{main}\, T(i,j) + d1\, T(i,j+1) + d2\, T(i+1,j) + d3\, T(i,j-1) + d4\, T(i-1,j) = \varnothing \quad (5)$$

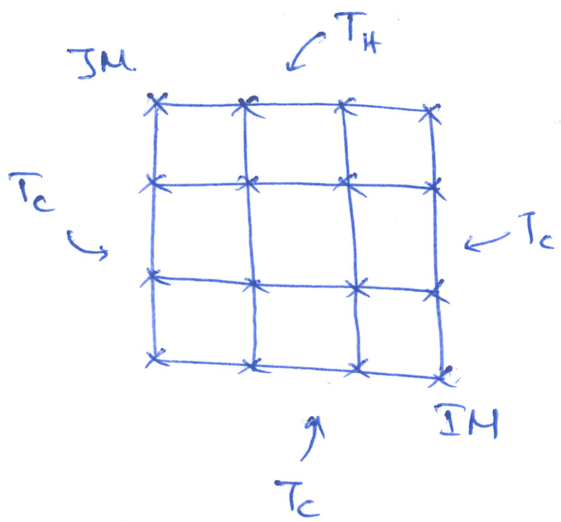
$$\hookrightarrow \underset{\sim}{A}\vec{x} = \vec{b} \rightarrow \underset{\sim}{A}\vec{T} = \vec{b}, \quad \vec{b} = \vec{\varnothing}$$

Residuals

$$\left.\begin{array}{l} \vec{R} = \vec{b} - \underset{\sim}{A}\vec{T} \\ \vec{b} = \vec{\varnothing} \end{array}\right\} \Rightarrow \vec{R} = -\underset{\sim}{A}\vec{T} \Rightarrow$$

$$\vec{R} = -\left[d_{main}\, T(i,j) + d1\, T(i,j+1) + d2\,(i+1,j) + d3\, T(i,j-1) + d4\,(i-1,j)\right] \quad (6)$$

Check with → Gauss-Seidel
→ Conjugate Gradient



$$\left.\begin{array}{l} T_H = 500\ °c \\ T_c = 100\ °c \end{array}\right\} \text{Diriclet BCs.}$$